

Chapter 3: part 1

EXERCISE 1. By finding the Wronskian of the following pairs of functions, show that they are linearly independent:

- (i) $x_1(t) = e^{k_1 t}$ and $x_2(t) = e^{k_2 t}$ with $k_1 \neq k_2$,
- (ii) $x_1(t) = e^{kt}$ and $x_2(t) = t e^{kt}$,
- (iii) $x_1(t) = e^{\rho t} \sin \omega t$ and $x_2(t) = e^{\rho t} \cos \omega t$.

EXERCISE 2. Show that the Wronskian for two solutions $x_1(t)$ and $x_2(t)$ of the second order differential equation $\frac{d^2 x}{dt^2} + p_1(t) \frac{dx}{dt} + p_2(t)x = 0$ satisfies

$$\frac{dW(t)}{dt} = -p_1(t)W(t).$$

Deduce either that $W(t) = 0$ for all t , or that $W(t) \neq 0$ for all t .

EXERCISE 3. We have seen that, if $x_1(t)$ and $x_2(t)$ are two solutions of a linear homogeneous second order differential equation, then they are linearly independent if and only if their Wronskian is non-zero. This is not true for general functions that are not the solutions of some differential equation.

- (i) Check that, if $f(t) = t^2|t|$, then $df/dt = 3t|t|$ (straightforward for $t \neq 0$; you will have to use the formal definition of the derivative at $t = 0$).
- (ii) Considering $f_1(t) = t^2|t|$ and $f_2(t) = t^3$ show that, although these two functions are linearly independent on $\mathbb{R}$, their Wronskian is identically zero.

EXERCISE 4. Find the general solution of the following differential equations, and then the solution satisfying the specified initial conditions.

- (i) $\ddot{x} - 3\dot{x} + 2x = 0$ with $x(0) = 2$ and $\dot{x}(0) = 6$;
- (ii) $y'' - 4y' + 4y = 0$ with $y(0) = 0$ and $y'(0) = 3$;
- (iii) $z'' - 4z' + 13z = 0$ with $z(0) = 7$ and $z'(0) = 42$;
- (iv) $\ddot{y} + \dot{y} - 6y = 0$ with $y(0) = -1$ and $y'(0) = 8$;
- (v) $\ddot{y} - 4\dot{y} = 0$ with $y(0) = 13$ and $\dot{y}(0) = 0$;
- (vi) $\ddot{\theta} + 4\theta = 0$ with $\theta(0) = 0$ and $\dot{\theta}(0) = 10$;
- (vii) $\ddot{y} + 2\dot{y} + 10y = 0$ with $y(0) = 3$ and $\dot{y}(0) = 0$;

- (viii) $2\ddot{z} + 7\dot{z} - 4z = 0$ with $z(0) = 0$ and $\dot{z}(0) = 9$;
- (ix) $\ddot{y} + 2\dot{y} + y = 0$ with $y(0) = 0$ and $\dot{y}(0) = -1$;
- (x) $\ddot{x} + 6\dot{x} + 10x = 0$ with $x(0) = 3$ and $\dot{x}(0) = 1$;
- (xi) $4\ddot{x} - 20\dot{x} + 21x = 0$ with $x(0) = -4$ and $\dot{x}(0) = -12$;
- (xii) $\ddot{y} + \dot{y} - 2y = 0$ with $y(0) = 4$ and $\dot{y}(0) = -4$;
- (xiii) $\ddot{y} - 4y = 0$ with $y(0) = 10$ and $\dot{y}(0) = 0$;
- (xiv) $y'' + 4y' + 4y = 0$ with $y(0) = 27$ and $y'(0) = -54$; and
- (xv) $\ddot{y} + \omega^2 y = 0$ with $y(0) = 0$ and $\dot{y}(0) = 1$.

EXERCISE 5. If the roots of the auxiliary equation are $k_1 > 0$ and $-k_2 < 0$ then the solution is $x(t) = Ae^{k_1 t} + Be^{-k_2 t}$. For most choices of initial conditions $x(0) = x_0$, $\dot{x}(0) = y_0$, we will have $x(t) \rightarrow \pm\infty$ as $t \rightarrow \infty$. However, there are some special initial conditions for which $x(t) \rightarrow 0$ as $t \rightarrow \infty$. Find the relationship between x_0 and y_0 that ensures this.

EXERCISE 6. Given an oscillating function of the form $x(t) = A \cos(\omega t) + B \sin(\omega t)$, show that it can be rewritten as a “single oscillation” $x(t) = M \cos(\omega t - \phi)$ with amplitude $M = (A^2 + B^2)^{1/2}$ and phase $\phi = \arctan(B/A)$.

EXERCISE 7. Consider a so-called simple pendulum whose shaft has length L and negligible mass, and a suspended mass m at its end. If the pivot is at the origin of coordinates, show that:

- (i) The gravitational potential energy $V(\theta) = mgL(1 - \cos \theta)$, where g is the acceleration of gravity and θ is the angle with the downward vertical.
- (ii) The total (kinetic plus potential) energy is $E = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos \theta)$.
- (iii) The equation of motion reads $\frac{d^2\theta}{dt^2} = -\frac{g}{L} \sin \theta$, which is obtained by requesting that the total energy does not change with time.
- (iv) By considering small oscillations around the vertical, the equation becomes $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$. Show that the general solution of this equation reads $\theta(t) = M \cos(\omega t - \phi)$ and relate the oscillation frequency ω with the system parameters. Does this frequency depend on the mass m ? Does it depend on the amplitude of oscillations?

EXERCISE 8. Consider the equation $m\ddot{x} + \mu\dot{x} + kx = 0$ for the deviation $x(t)$ of a point particle with mass m from its equilibrium position, subject to an elastic force (with elastic constant k) and a drag force (with damping parameter μ). Discuss how the form of the solutions changes, depending on the value of the damping μ relative to the so-called critical damping $\mu_c = 2\sqrt{mk}$.

EXERCISE 9. A chain of length $\ell = 5\text{m}$ and mass m , with uniform mass density, is sitting on a table. If the chain is initially at rest and 1m of the chain hangs from the edge of the table, apply Newton's second law to calculate the time t_* when the whole chain falls off the table, neglecting any friction.

Hint: Formulate a second order ODE for the variable $x(t)$, which is the length of the portion of chain that is overhanging, as a function of time.

EXERCISE 10. A parachutist of mass m Kg jumps from a plane with initial velocity 0. Because of the parachute, there is a friction force that avoids free fall; this friction turns out to be proportional (with constant of proportionality $c > 0$) to the velocity. Write down an ODE to describe the motion of the parachutist, and determine the terminal speed (when $t \rightarrow \infty$). Compare this speed with that of free fall.

EXERCISE 11. We consider the following second order ODE (Euler equation)

$$x^2 y''(x) + p x y'(x) + q y(x) = g(x),$$

where p, q are constants.

- i) By making the change of variable $x = e^z$ and using the chain rule, show that the ODE is transformed into one with constant coefficients in the variable z :

$$\frac{d^2 y}{dz^2} + (p-1) \frac{dy}{dz} + y = g(e^z).$$

- ii) Take $p = 3$, $q = -3$ and $g(x) = 4x$, make the previous transformation and compute the general solution.
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